

The prevalence of bacterial vaginosis in the United States, 2001-2004; associations with symptoms, sexual behaviors, and reproductive health.



OBJECTIVES: Bacterial vaginosis (BV), a disturbance of vaginal microflora, is a common cause of vaginal symptoms and is associated with an increased risk of acquisition of sexually transmitted infections, HIV, and with adverse pregnancy outcomes. We determined prevalence and associations with BV among a representative sample of women of reproductive age in the United States. **STUDY DESIGN:** Women aged 14-49 years participating in the National Health and Nutrition Examination Survey 2001-2004 were asked to submit a self-collected vaginal swab for Gram staining. BV, determined using Nugent's score, was defined as a score of 7-10. **RESULTS:** The prevalence of BV was 29.2% (95% confidence interval 27.2%-31.3%) corresponding to 21 million women with BV; only 15.7% of the women with BV reported vaginal symptoms. Prevalence was 51.4% among non-Hispanic blacks, 31.9% among Mexican Americans, and 23.2% among non-Hispanic whites ($P < 0.01$ for each comparison). Although BV was also associated with poverty ($P < 0.01$), smoking ($P < 0.05$), increasing body mass index ($\chi^2 P < 0.0001$ for trend), and having had a female sex partner ($P < 0.005$), in the multivariate model, BV only remained positively associated with race/ethnicity, increasing lifetime sex partners ($\chi^2 P < 0.001$ for trend), increasing douching frequency ($\chi^2 P$ for trend < 0.001), low educational attainment ($P < 0.01$), and inversely associated with current use of oral contraceptive pills ($P < 0.005$). **CONCLUSION:** BV is a common condition; 84% of women with BV did not report symptoms. Because BV increases the risk of acquiring sexually transmitted infections, BV could contribute to racial disparities in these infections.